

The Dimension of Nonterminating Resampling Computations

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Abstract

A randomized algorithm may terminate almost surely even though exceptional random tapes make it run forever. This paper studies the survival tail, the Kolmogorov complexity of one such tape, and the Hausdorff dimension of all of them. For each $s > 0$ at which the powered repair matrices commute, the main theorem bounds $\sum_w P[w]^s$ over surviving prefixes w , uniformly over deterministic nonanticipating selectors. The case $s = 1$ controls termination; the full family gives weak-source and dimension bounds.

The source powers contain information absent even from the ordinary repair kernel and the complete stopping-time law. Under one common finite tape source, two overlapping disagreement-repair rules on a four-vertex path have the same ordinary kernels and the same stopping-time law for every selector, yet their nontermination dimensions can be arbitrarily close to zero and one. At one common source-power level, the same dominated tape source makes one rule run forever but gives the other an exponential stopping tail. The separation is caused by action labels that produce the same state transition and are therefore invisible at power one. For bounded-dependence k -SAT, conditional block min-entropy above the trace-growth threshold gives exponential termination, and the effective dimension of an individual infinite run is bounded by the trace growth induced by the clauses repaired infinitely often. Tree formulas asymptotically attain the maximum-degree dimension and global source bounds, while clique formulas attain the graph-specific one-step threshold in the stated regime. An exact backward likelihood identity complements these setwise results with tail and coding bounds for each run.

1 Introduction

A randomized algorithm is deterministic once its random tape is fixed. Although the algorithm may terminate with probability one, some tapes may make it run forever. Let $\mathbb{P}$ denote the source law of the tape and put

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where τ is the number of repair steps before stopping. The tail $\mathbb{P}\{\tau \geq T\}$ measures how often the computation survives T steps. When the following extended limit exists, the *survival exponent* is

$$\alpha = - \lim_{T \rightarrow \infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

This is the base-two form of the usual escape rate (Keller and Liverani, 2009).

The survival exponent does not describe the exceptional tapes themselves. For a computable process with a computable tape law P , $\mathcal{N}$ is effectively closed because termination, when it occurs, is determined by a finite prefix. Thus a P -null $\mathcal{N}$ contains no P -Martin-Löf-random tape, but it may still have positive or full Hausdorff dimension and contain tapes of maximal effective strong dimension. Effective dimension measures the algorithmic randomness of one tape; source-relative Hausdorff dimension measures the size of the whole exceptional set.

The Lovász local lemma has a different scope. It gives conditions under which all bad events can be avoided, and its algorithmic forms give termination or tail guarantees from event probabilities and dependency structure (Erdős and Lovász, 1975; Moser and Tardos, 2010; Haeupler and Harris, 2017). The present paper fixes the resampling computation and asks for its survival rate and for the complexity and geometry of its exceptional tapes. These are process-specific questions, rather than a stronger universal local-lemma criterion. The local lemma is valuable because it supplies efficiently checkable quantitative bounds without constructing the full assignment chain.

The motivation is pointwise. In statistical learning, the learned rule $\hat{h}(S)$ depends on the sample S , so a bound for a fixed rule cannot simply be evaluated at $\hat{h}(S)$; uniform and localized empirical-process methods control the data-dependent region in which it lies (Bartlett et al., 2005; Koltchinskii, 2006; Xu and Zeevi, 2025). A resampling computation creates an analogous dependence between a tape and the repair history generated from it: survival for T repairs restricts tape space to a nested family of live cylinders. Recent work studies complexity at one trained network or one index of a random field (Li and Xu, 2026; Xu, 2026b); here the pointwise object is one random tape. This is an analogy, not a reduction to a learning bound: it isolates the common question of how much complexity remains after data-dependent localization. The search for a smaller process state that preserves future source-power sums likewise parallels Bellman-sufficient representations (Xu, 2026a). Corollary 4.2 makes this localization precise: the effective strong dimension of an infinite k -SAT run is bounded by the trace growth of the subgraph induced by the clauses repaired infinitely often.

For a computable full-support source $P = \pi^{\mathbb{N}}$, let $\omega_{1:n}$ denote the first n source symbols, and let

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}]$$

be their self-information. If $K(w)$ is prefix-free Kolmogorov complexity, the effective strong P -dimension of one tape is

$$\text{Dim}_{\text{eff},P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \quad (1)$$

This is the standard source-relative notion (Staiger, 1993; Athreya et al., 2007; Lutz, 2011). A value near one means that the tape retains nearly the full algorithmic information of the source; a smaller value means that continued execution has imposed additional structure. The source-relative Hausdorff dimension $\dim_H^P E$ of a tape set E is defined by assigning diameter $P[w]$ to the cylinder $[w]$ (Billingsley, 1965; Lutz, 2011). Section B.1 gives the corresponding source tree when tape consumption is adaptive.

For the prefix-free family $\mathcal{L}_T$ of minimal tape prefixes that remain live for T repairs, define

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s. \quad (2)$$

This Rényi sum is the central object. At $s = 1$ it is the survival probability; its behavior across s controls Hausdorff dimension through the mass-distribution principle and Frostman’s lemma (Rényi, 1961; Falconer, 2014). It also turns global source-power domination—and, for a uniform reference, a min-entropy bound—into a running-time bound for the original algorithm.

1.1 Results

Theorem 2.3 is a source-power extension of matrix-commutative local repair. For each flaw f and power $s > 0$, the matrix $K_{f,s}$ sums powered action probabilities over state transitions. When the matrices for nonadjacent flaws commute, the theorem bounds $Z_T^\Lambda(s)$ by the Cartier–Foata/Shearer

trace series, term by term and uniformly over every deterministic nonanticipating rule Λ . The direct full-prefix induction permits colliding actions and rules that remember their labels. At $s = 1$ it gives familiar termination estimates; varying s gives weak-source robustness and bounds on Hausdorff and effective dimension.

The principal exact application shows why the full source-power family is needed. In disagreement repair on a connected graph, Hamming weight reduces every $Z_T(s)$ to one tridiagonal matrix, independently of the graph and the repair order. On P_4 the repairs overlap and nonadjacent repairs commute. Under one common finite tape source, two rules can have identical ordinary repair kernels and the same complete stopping-time law for every selector, while their sharp source-domination thresholds and nontermination dimensions are arbitrarily close to zero and one (Theorem 3.1 and Corollary 3.2). Thus even complete running-time information—or the ordinary transition kernel—need not determine the geometry of exceptional tapes or robustness to weak randomness.

For k -SAT, conditional block min-entropy above $\log_2 \kappa_D$ gives an exponential tail, where κ_D is the trace-growth rate of the clause-overlap graph. Corollary 4.2 gives a pointwise upper bound in terms of the clauses repaired infinitely often. Tree formulas asymptotically attain the maximum-degree dimension and global source bounds; clique formulas attain the graph-specific one-step threshold in the stated regime. Further consequences include fresh-bit and Santha–Vazirani guarantees and rainbow perfect matching. Complementing these setwise results, Theorem 5.1 reads one finite run backward and expresses its information surplus as an exact likelihood ratio, yielding a tail bound and a description-length saving for every surviving prefix.

2 Source-power local repair

This section bounds the complete source-power sum of an overlapping repair process without constructing its full assignment chain. The input is a family of powered transition matrices and local scalar bounds on their growth. The conclusion is uniform over the order in which present flaws are selected.

Let Ω be a finite state space and let $F_\sigma \subseteq F$ be the flaws present at σ . Addressing $f \in F_\sigma$ chooses an action $a \in \mathcal{A}_f(\sigma)$ with probability $\rho_f(a \mid \sigma) > 0$ and moves to $\Phi_f(\sigma, a)$. The action sets are finite and satisfy

$$\sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma) = 1.$$

Distinct actions may lead to the same terminal state. A deterministic nonanticipating selector may use the complete observed past, including past action labels, but not future randomness.

Let $\sim$ be a symmetric relation on distinct flaws and define the inclusive neighborhood

$$\Gamma(f) = \{f\} \cup \{g \in F : g \sim f\}.$$

In local-repair applications, $\sim$ is also chosen to satisfy the potential-causality condition

$$F_{\Phi_f(\sigma, a)} \subseteq (F_\sigma \setminus \{f\}) \cup \Gamma(f). \quad (3)$$

This is Kolmogorov’s convention after adding a self-loop at every flaw. The matrix theorem itself uses only powered commutation; potential causality explains why D is a local dependency graph. Write $\text{Ind}(S)$ for the subsets of S whose distinct elements are pairwise nonadjacent.

For $s > 0$, define the powered action kernel

$$K_{f,s}(\sigma, \tau) = \mathbf{1}\{f \in F_\sigma\} \sum_{\substack{a \in \mathcal{A}_f(\sigma) \\ \Phi_f(\sigma, a) = \tau}} \rho_f(a \mid \sigma)^s. \quad (4)$$

The kernel aggregates actions with the same endpoints, but for $s \neq 1$ its entries retain the powered probabilities of the individual actions and need not be determined by the ordinary transition matrix $K_{f,1}$. The process is s -commutative if

$$K_{f,s}K_{g,s} = K_{g,s}K_{f,s} \quad (f \neq g, f \not\sim g). \quad (5)$$

Two familiar conditions imply this matrix identity. First, suppose that for nonadjacent f, g , action-labelled paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

are in product-preserving bijection with paths

$$\sigma \xrightarrow{g,b'} \sigma'' \xrightarrow{f,a'} \tau,$$

so that

$$\rho_f(a \mid \sigma)\rho_g(b \mid \sigma') = \rho_g(b' \mid \sigma)\rho_f(a' \mid \sigma''). \quad (6)$$

Second, suppose that each enabled instance of flaw f has the same number M_f of equiprobable actions.

Lemma 2.1 (Two routes to powered commutation). *The product-preserving diamond property implies (5) for every $s > 0$. Alternatively, if each enabled f has exactly M_f actions of probability $1/M_f$, then*

$$K_{f,s} = M_f^{1-s} K_{f,1}. \quad (7)$$

Consequently ordinary matrix commutation at $s = 1$ implies (5) for every $s > 0$ in this fixed-arity case. Different action labels may lead to the same terminal state.

The proof is in Appendix B.3.

Ordinary matrix commutation at $s = 1$ need not survive powering. For example,

$$A_0 = \begin{pmatrix} 4/5 & 1/5 \\ 2/5 & 3/5 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 7/10 & 3/10 \\ 3/5 & 2/5 \end{pmatrix}$$

commute, whereas their entrywise square roots do not. Thus the full source-power family requires additional structure beyond ordinary matrix commutation; Lemma 2.1 gives two sufficient conditions.

Fix an initial law u on Ω . The initial source symbol is taken to be the initial state, and its powered mass vector is

$$b_s(\sigma) = u(\sigma)^s.$$

If instead an initial symbol α of probability $q(\alpha)$ produces the state $\iota(\alpha)$, replace this by

$$b_s(\sigma) = \sum_{\alpha: \iota(\alpha)=\sigma} q(\alpha)^s.$$

One later source symbol is one complete oracle action. Let $\mathcal{L}_T^\Lambda$ be the prefix-free family of minimal tape prefixes that complete T repairs under selector Λ , and set

$$Z_T^\Lambda(s) = \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s, \quad (8)$$

where $P_{\text{tape}}[w]$ includes the initial source symbol. With this convention, $Z_T^\Lambda(1) = \mathbb{P}\{\tau \geq T\}$ when τ is the number of completed repairs.

Write $f \simeq g$ when $f = g$ or $f \sim g$. For a flaw word $\ell = (f_1, \dots, f_T)$, its *full history DAG* has vertex set $[T]$ and an edge $i \rightarrow j$ exactly when $i < j$ and $f_i \simeq f_j$. For such a DAG H , put

$$K_{\emptyset, s} = I, \quad K_{H, s} = K_{L(x), s} K_{H-x, s},$$

where x is any source and $L(x)$ is its label. Under (5), this definition is independent of the chosen source.

Two full history DAGs are identified when they are related by a label-preserving directed-graph isomorphism.

Choose a strictly positive probability vector v_s on Ω and define

$$\lambda_f(s) = \max_{\tau \in \Omega} \frac{\sum_{\sigma \in \Omega} v_s(\sigma) K_{f, s}(\sigma, \tau)}{v_s(\tau)}, \quad (9)$$

$$\gamma_s = \max_{\sigma \in \Omega} \frac{b_s(\sigma)}{v_s(\sigma)}. \quad (10)$$

Let D be the graph on F with edges $f \sim g$ and let $\mathcal{M}(D)$ be the trace monoid in which nonadjacent flaws commute. For a trace h , write $N_f(h)$ for the multiplicity of f and $|h| = \sum_f N_f(h)$. Define

$$a_T(s) = \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_{f \in F} \lambda_f(s)^{N_f(h)}, \quad (11)$$

$$P_D(\mathbf{z}) = \sum_{I \in \text{Ind}(F)} (-1)^{|I|} \prod_{f \in I} z_f. \quad (12)$$

The open Shearer region consists of the vectors $\mathbf{z} \geq 0$ for which $P_D(\mathbf{y}) > 0$ for every $0 \leq \mathbf{y} \leq \mathbf{z}$. The Cartier–Foata identity gives

$$\sum_{T \geq 0} a_T(s) z^T = \frac{1}{P_D(z\boldsymbol{\lambda}(s))} \quad (13)$$

as a formal power series, where $\boldsymbol{\lambda}(s) = (\lambda_f(s))_{f \in F}$.

For two formal series $F(z) = \sum_{T \geq 0} a_T z^T$ and $G(z) = \sum_{T \geq 0} b_T z^T$, write $F \preceq_{\text{term}} G$ when $a_T \leq b_T$ for every T .

At $s = 1$, Harris et al. (2025, Lemma 3.5) bound the probability that a prescribed history DAG appears during a run. Their induction allows the current repair to be absent from that DAG and uses substochasticity in this case. Here H records the entire next T -repair prefix, so the current repair must label a source of H . The omission case disappears.

This distinction should not be overstated. At a fixed power, scalar normalization makes every powered kernel sub-Markov. For a rule determined by the state history, a clock and a final flaw adjacent to every original flaw reduce a length- T prefix to the appearance event analyzed by Harris et al. (2025); the immediate killed-process version of their lemma then recovers the corresponding special case below after the scalar factors are restored. The direct induction is useful because it needs no auxiliary process and, more importantly, permits colliding actions to be distinguished by a rule that observes their labels; encoding those labels into a state-only process would require enlarging the state space.

Lemma 2.2 (Exact-prefix matrix bound). *Fix $s > 0$ for which (5) holds. For a full history DAG H with T vertices, let $\mathcal{L}_T^\Lambda(H)$ be the live prefixes whose first T repair labels induce H . Every deterministic nonanticipating selector satisfies*

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq b_s^\top K_{H,s} \mathbf{1}. \quad (14)$$

No injectivity of the action-to-state map is required.

For positive numbers $(\eta_f)_{f \in F}$, write $\eta(I) = \prod_{f \in I} \eta_f$ and put

$$\theta_s = \max_{f \in F} \frac{\lambda_f(s)}{\eta_f} \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I). \quad (15)$$

Theorem 2.3 (Powered matrix-commutative local lemma). *Fix $s > 0$ and suppose that the powered kernels satisfy (5). Then every deterministic nonanticipating selector Λ satisfies*

$$Z_T^\Lambda(s) \leq \gamma_s a_T(s) \quad (T \geq 0), \quad (16)$$

or, equivalently,

$$\sum_{T \geq 0} Z_T^\Lambda(s) z^T \preceq_{\text{term}} \frac{\gamma_s}{P_D(z\Lambda(s))}. \quad (17)$$

If $q\Lambda(s)$ lies in the open Shearer region for some $q > 1$, then

$$Z_T^\Lambda(s) \leq \frac{\gamma_s}{P_D(q\Lambda(s))} q^{-T}. \quad (18)$$

If instead $\theta_s < 1$, then

$$Z_T^\Lambda(s) \leq \gamma_s \left(\sum_{R \in \text{Ind}(F)} \eta(R) \right) \theta_s^T \quad (T \geq 0). \quad (19)$$

All quantities on the right are independent of the selector.

Proof idea. Fix a full T -step history DAG H and a past ending at state σ . Write $M_{\mathfrak{h}}(H)$ for the powered mass of continuations whose next T repairs induce H . If the rule next selects f and no source of H has label f , the powered mass of the corresponding continuations is zero. Otherwise that source x is unique. Conditioning on the next action and applying induction to $H - x$ gives

$$M_{\mathfrak{h}}(H) \leq e_\sigma^\top K_{f,s} K_{H-x,s} \mathbf{1}.$$

Matrices attached to different sources commute, so $K_{H,s}$ does not depend on the order in which sources are removed. Summing over the initial symbol yields

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq b_s^\top K_{H,s} \mathbf{1}.$$

Because H records the entire repair prefix, there is no omission case; actions with the same endpoints are added in $K_{f,s}$. The positive vector v_s bounds the last display by $\gamma_s \prod_{x \in H} \lambda_{L(x)}(s)$. Full history DAGs, up to label-preserving isomorphism, are traces. Summing over traces and applying the

Cartier–Foata identity proves the term-by-term bound. The appendix gives the full induction and the Shearer and cluster estimates.

The term-by-term trace bound is the sharpest of these statements. The Shearer condition turns it into an exponential estimate, while the cluster expression is a convenient local sufficient condition. At $s = 1$ they recover established commutative-LLL bounds. The new use is to control the full source-power family, and hence weak randomness and exceptional-tape dimension, by the same matrices.

Whenever either strict condition in Theorem 2.3 holds, choose constants such that

$$Z_T^\Lambda(s) \leq C_s \vartheta_s^T, \quad C_s < \infty, \quad 0 < \vartheta_s < 1, \quad (20)$$

using the corresponding constants from (18) or (19). If the rate supplied by the theorem is zero, the same estimate remains valid after replacing it by any fixed number in $(0, 1)$.

Corollary 2.4 (Dimension and pointwise coding). *Assume (20) and $\rho_{\max} = \max_{\sigma, f, a} \rho_f(a \mid \sigma) < 1$. Then*

$$\dim_H^{P_{\text{tape}}} \mathcal{N} \leq s. \quad (21)$$

Suppose in addition that s , the process, the selector, and computable numbers $\widehat{C}_s \geq C_s$ and $0 < \widehat{\vartheta}_s < 1$ with $\widehat{\vartheta}_s \geq \vartheta_s$ are computable. Include fixed descriptions of these objects in instance. Every $w \in \mathcal{L}_T^\Lambda$ then satisfies

$$K(w \mid T, \text{instance}) \leq s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1), \quad (22)$$

and every $\omega \in \mathcal{N}$ has $\text{Dim}_{\text{eff}, P}(\omega) \leq s$ in this action-outcome source tree.

The metric in this corollary is tied to complete oracle actions. A binary random-bit implementation gives the same conclusion when it is related to this tree by a computable prefix-free sampler with bounded distortion; no invariance under arbitrary implementations is asserted.

Corollary 2.5 (Dependent repair choices). *Let ν be a possibly history-dependent law that uses only actions available at the corresponding reference history. If a constant $D < \infty$ satisfies*

$$\nu[w] \leq D P_{\text{tape}}[w]^s \quad (w \in \mathcal{L}_T^\Lambda, T \geq 0), \quad (23)$$

then

$$\mathbb{P}_\nu\{\tau \geq T\} \leq D C_s \vartheta_s^T. \quad (24)$$

When $0 < s \leq 1$ and the reference oracle is uniform on the same M_t actions available at step t , the explicit condition

$$\max_a \mathbb{P}_\nu\{A_t = a \mid \mathcal{H}_{t-1}\} \leq M_t^{-s}$$

for every feasible positive-probability history implies (23) with $D = 1$ when ν and P_{tape} also use the same initial law. The repair choices may be correlated across time, but may alter neither the selector nor the valid action sets. For a different initial law, or for $s > 1$, the corresponding initial likelihood ratio is absorbed into D when it is bounded.

3 Exact overlapping disagreement repair

The next result is an exact, nonextremal application in which local repairs overlap and the induced transition depends on the current state. It also shows that the complete stopping-time law does not determine the dimension of the tapes on which the computation never stops.

Let $G = (V, E)$ be a connected graph with $N = |V| \geq 2$, and give every edge a fixed orientation. A state is $x \in \{0, 1\}^V$, and an edge is flawed when its endpoints disagree. Let $\mathcal{A}$ be a finite action alphabet with a full-support law π , and write $P = \pi^{\mathbb{N}}$ for the action-tape law (the initial assignment is fixed). Partition the alphabet into four nonempty classes $\mathcal{A}_-, \mathcal{A}_p, \mathcal{A}_r, \mathcal{A}_+$. When a selected oriented edge reads $z \in \{01, 10\}$, a symbol in these four classes replaces z , respectively, by 00, by z , by the reversed word $\bar{z}$, or by 11. Thus several action labels may produce the same state transition. The tape consists of independent draws from π , and any deterministic nonanticipating rule may choose the next flawed edge, even using the earlier action labels.

For $s > 0$, define the three source-power masses

$$D_s = \sum_{a \in \mathcal{A}_-} \pi(a)^s, \quad N_s = \sum_{a \in \mathcal{A}_p \cup \mathcal{A}_r} \pi(a)^s, \quad U_s = \sum_{a \in \mathcal{A}_+} \pi(a)^s. \quad (25)$$

This is a four-outcome extension of discordant voting (Cooper et al., 2018). Conditional on a selected discordant edge, the oblivious voting update is the boundary case with mass $1/2$ on each of 00 and 11: the selected edge becomes monochromatic. Here the selected edge may remain discordant, either unchanged or reversed, and the edge selector may be any deterministic nonanticipating rule. Cooper et al. (2018) study expected consensus time; the result below determines the complete stopping-time law and source-power geometry for the extended rule.

For P_4 , Hamming weight collapses the fourteen live assignments to three states:

$$\begin{aligned} P_4 : \quad & \{x \in \{0, 1\}^4 : x \text{ is live}\} \xrightarrow{M(x)=\sum_v x_v} \{1, 2, 3\}, \\ 0_{\text{stop}} & \xleftarrow{D_s} 1 \xrightleftharpoons[D_s]{U_s} 2 \xrightleftharpoons[D_s]{U_s} 3 \xrightarrow{U_s} 4_{\text{stop}}, \quad R_s(k, k) = N_s. \end{aligned} \quad (26)$$

Theorem 3.1 (Exact disagreement repair with colliding actions). *Start from a fixed live assignment with k_0 ones. For $s > 0$, let R_s be the $(N-1) \times (N-1)$ tridiagonal matrix*

$$R_s(k, k-1) = D_s, \quad R_s(k, k) = N_s, \quad R_s(k, k+1) = U_s, \quad (27)$$

where entries outside $\{1, \dots, N-1\}$ are omitted. For every deterministic nonanticipating selector,

$$Z_T(s) = e_{k_0}^T R_s^T \mathbf{1}. \quad (28)$$

In particular, for fixed N and k_0 , the complete stopping-time law is independent of G and of the selector. With $c_N = \cos(\pi/N)$, the exact source-power rate is

$$\varrho_s = \rho(R_s) = N_s + 2(D_s U_s)^{1/2} c_N. \quad (29)$$

The process terminates almost surely with an exponential tail, and

$$\dim_H^P \mathcal{N} = \Delta, \quad N_\Delta + 2(D_\Delta U_\Delta)^{1/2} c_N = 1, \quad (30)$$

where the equation has a unique solution $\Delta \in (0, 1)$. If π , the repair map, and the selector are computable, then

$$\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \Delta, \quad (31)$$

and the supremum is attained.

The same Δ is the exact threshold for global source-power domination. If a possibly dependent action source ν satisfies

$$\nu[w] \leq DP[w]^s \quad (w \in \mathcal{L}_T, T \geq 0) \quad (32)$$

for some $D < \infty$ and $s > \Delta$, then

$$\mathbb{P}_\nu\{\tau \geq T\} \leq De_{k_0}^\top R_s^T \mathbf{1} = O(\varrho_s^T). \quad (33)$$

At $s = \Delta$, there is a law ν_Δ on the same action tree, supported on $\mathcal{N}$, and a constant $D_\Delta < \infty$ for which $\nu_\Delta[w] \leq D_\Delta P[w]^\Delta$ for every prefix w ; since $P[w] \leq 1$, the same law is dominated at every $0 < s \leq \Delta$.

Proof sketch. Every flawed edge contains exactly one vertex of value one. An action therefore changes Hamming weight by -1 , 0 , or $+1$, with total source-power mass D_s , N_s , or U_s , independently of the selected edge. Summing prefixes by their current weight gives the recurrence with R_s and proves (28). A diagonal similarity turns R_s into a symmetric tridiagonal matrix with off-diagonal entry $(D_s U_s)^{1/2}$, yielding (29). The finite-state pressure formula and its Perron–Doob law then give the two dimensions and the sharp source-domination threshold.

The proof, including the critical law attaining the threshold and both dimensions, is in Appendix B.4.

Corollary 3.2 (A common-source same-kernel separation). *For every $\varepsilon > 0$, there exist a computable finite source P and two disagreement-repair maps H and L on P_4 with the following properties. Their ordinary repair matrices agree for every flaw and state, $K_{f,1}^H = K_{f,1}^L$. Regardless of which deterministic nonanticipating selector is used in either process, including one that remembers the action labels, the stopping law depends only on the initial Hamming weight and is common to H and L . Nevertheless, for every common live initial assignment and every such pair of selectors, their nontermination languages, measured in the same source metric, satisfy*

$$\dim_H^P \mathcal{N}_L = \Delta_L < \varepsilon, \quad \dim_H^P \mathcal{N}_H = \Delta_H > 1 - \varepsilon. \quad (34)$$

For computable selectors, Δ_L and Δ_H are also the attained maximal effective strong dimensions.

Moreover, after fixing the initial assignment and the selectors, choose any s_0 with $\Delta_L < s_0 < \Delta_H$. There exists a single tape source ν_H , satisfying $\nu_H[w] \leq DP[w]^{s_0}$ for every finite action word w , such that H never terminates under ν_H , whereas L , driven by the same ν_H , has an exponential stopping tail. Thus the two rules have opposite sharp behavior under the same imperfect source despite having the same nominal transition matrices.

Here is the construction. Choose an integer $m \geq 2$, a number $0 < t < 1/2$, and put $q = 1/2 - t$. The common alphabet is $\{x_1, \dots, x_m, y, u, v\}$, with

$$\pi(x_i) = q/m, \quad \pi(y) = q, \quad \pi(u) = \pi(v) = t. \quad (35)$$

For a selected edge reading z , use the two maps

	$x_1, \dots, x_m$	y	u	v
H	z	00	11	$\bar{z}$
L	00	z	11	$\bar{z}$.

(36)

Both maps therefore assign ordinary masses (q, q, t, t) to $(00, z, 11, \bar{z})$, so their $K_{f,1}$ matrices agree. The labels $x_1, \dots, x_m$, however, collide at one transition. The two exact source-power rates on P_4 are

$$\varrho_H(s) = m^{1-s}q^s + t^s + \sqrt{2}(qt)^{s/2}, \quad (37)$$

$$\varrho_L(s) = q^s + t^s + \sqrt{2}m^{(1-s)/2}(qt)^{s/2}. \quad (38)$$

As $t \downarrow 0$, their roots converge, respectively, to

$$\frac{\log m}{\log(2m)} \quad \text{and} \quad 0. \quad (39)$$

Taking m large and then t small proves (34). The languages themselves differ: from Hamming weight one, x_1^∞ survives under H and stops under L , while y^∞ does the reverse. Appendix B.4 gives the complete proof.

The role of collisions has a simple information-theoretic form. If $r = K_{f,1}(\sigma, \tau) > 0$ and χ is the conditional law of the action label among those producing $\sigma \rightarrow \tau$, then

$$K_{f,s}(\sigma, \tau) = r^s \sum_a \chi(a)^s = r^s 2^{(1-s)H_s(\chi)}. \quad (40)$$

Here H_s is base-two Rényi entropy, with its continuous interpretation at $s = 1$. Thus $K_{f,1}$ records the total transition mass, whereas the source powers also record the Rényi entropy hidden behind that transition. Under action recoverability every fiber is a singleton and this extra term vanishes. In (36), an m -way collision moves between the neutral and downward transitions while all power-one data remain fixed.

Adjacent edge flaws on P_4 overlap and can occur simultaneously, whereas the two end-edge repairs have disjoint scopes and commute at every source power. Thus commutation is nonvacuous, but the example is genuinely nonextremal and overlapping. The example shows that even the entire ordinary repair kernel can lose essentially all information about weak-source robustness and nontermination dimension.

4 Weak randomness, localization, and sharpness

For a variable-model event A , let $v(A)$ be its resampled coordinates and let $\pi_{v(A)}$ be their product law. Define its local source-power mass by

$$r_A(s) = \sum_{b \in A} \pi_{v(A)}(b)^s. \quad (41)$$

Here A is identified with its local configurations on $v(A)$. Taking $v_s(\sigma)$ proportional to $\pi(\sigma)^s$ in (9) gives the exact local charge

$$\lambda_A(s) = r_A(s). \quad (42)$$

Thus Theorem 2.3 can be checked from event scopes, local source masses, and the causality graph, without enumerating all assignments. Replacing the independent-set sum in (15) by $\prod_{B \in \Gamma(A)} (1 + \eta_B)$ gives a simpler polynomial-time sufficient test, though generally a weaker one. The independent-set refinement is the standard cluster-expansion form (Bissacot et al., 2011; Pegden, 2014); the use made here of the full source-power family is separate.

For a finite graph D , let

$$\kappa_D = \limsup_{T \rightarrow \infty} |\{h \in \mathcal{M}(D) : |h| = T\}|^{1/T}. \quad (43)$$

Equivalently, $\kappa_D = R_D^{-1}$, where R_D is the first positive zero of $z \mapsto P_D(z\mathbf{1})$.

Theorem 4.1 (Graph-specific min-entropy threshold for k -SAT). *Let Φ be a k -CNF with clause-overlap graph D , start from a fixed assignment, and use any deterministic nonanticipating selector. Let P be the law of independent uniform k -bit redraws. If a possibly history-dependent redraw law ν satisfies, for some $D_0 < \infty$ and $s > 0$,*

$$\nu[w] \leq D_0 P[w]^s \quad (44)$$

for every finite live redraw word w , then, for every $\xi > \kappa_D$,

$$\mathbb{P}_\nu\{\tau \geq T\} \leq \frac{D_0 2^n}{P_D(\xi^{-1}\mathbf{1})} (\xi 2^{-ks})^T. \quad (45)$$

In particular, the tail is exponential whenever

$$s > \frac{\log_2 \kappa_D}{k}. \quad (46)$$

If $h > 0$ and each redraw has conditional min-entropy at least h , then (45) holds with $D_0 = 1$ and $ks = h$. Consequently $h > \log_2 \kappa_D$ implies exponential termination. For the uniform reference law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 \kappa_D}{k} \right\}. \quad (47)$$

If the selector is computable, the same bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$.

If D has maximum degree at most $d \geq 2$, then

$$\kappa_D \leq c_d, \quad c_d = \frac{d^d}{(d-1)^{d-1}}. \quad (48)$$

Thus $h > \log_2 c_d$ is a selector-uniform sufficient condition, and the right side of (47) is at most $\min\{1, \log_2(c_d)/k\}$.

The one-step min-entropy hypothesis implies (44) by the chain rule. The converse can fail: a globally dominated source may move randomness between different times. The tree family below makes the global threshold and the dimension bound sharp, but does not by itself prove sharpness of the one-step conditional threshold $\log_2 c_d$.

For an infinite run ω , let

$$\text{Rec}(\omega) = \{A : A \text{ is repaired infinitely often on } \omega\}. \quad (49)$$

This set is nonempty because there are finitely many clauses.

Corollary 4.2 (Pointwise localization to recurrent clauses). *For a computable k -SAT repair process and every infinite run ω ,*

$$\text{Dim}_{\text{eff}, P}(\omega) \leq \min \left\{ 1, \frac{\log_2 \kappa_{D[\text{Rec}(\omega)]}}{k} \right\}. \quad (50)$$

For each nonempty $R \subseteq F$, the same right side with $D[R]$ bounds the Hausdorff dimension of the tapes satisfying $\text{Rec}(\omega) = R$.

This is the precise localization statement behind the motivation in Section 1: clauses repaired only finitely often affect a finite prefix but not the asymptotic complexity of that tape.

For $d \geq 2$, define

$$A_d = 1 + \frac{d^d}{(d-1)^{d-1}}. \quad (51)$$

The graph-specific theorem is sharper but involves P_D . The following local relaxation has explicit constants and, relative to the worst-case max-degree bound c_d , loses at most $\log_2(A_d/c_d) \leq \log_2(5/4)$ bits per repair.

Theorem 4.3 (An explicit max-degree tail for k -SAT). *Let Φ be a k -CNF on n variables with m clauses, each containing k distinct variables and overlapping at most $d \geq 2$ other clauses. Start from an arbitrary assignment and use any deterministic nonanticipating rule to select a violated clause. Suppose that, at every feasible history, the next k -bit redraw B_t has worst-case conditional min-entropy at least h , in the pointwise sense*

$$\max_b \mathbb{P}\{B_t = b \mid \mathcal{H}_{t-1}\} \leq 2^{-h}. \quad (52)$$

If $\varepsilon = h - \log_2 A_d > 0$, then

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left(\frac{d}{d-1} \right)^m 2^{-\varepsilon T}. \quad (53)$$

For independent uniform redraws, this becomes

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left(\frac{d}{d-1} \right)^m \left(\frac{A_d}{2^k} \right)^T. \quad (54)$$

Writing P for the uniform independent repair-tape law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 A_d}{k} \right\}. \quad (55)$$

If the selector is computable, the same upper bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$. The tail and Hausdorff-dimension conclusions are uniform over every deterministic nonanticipating selector and every history-dependent redraw law satisfying (52).

Corollary 4.4 (Fresh fair bits). *Before repair t , let the past determine an injective map*

$$E_t : \{0, 1\}^r \longrightarrow \{0, 1\}^k, \quad 1 \leq r \leq k,$$

then read r fresh independent fair bits U_t and redraw the selected clause to $E_t(U_t)$. If $\varepsilon_r = r - \log_2 A_d > 0$, then (53) holds with $\varepsilon = \varepsilon_r$, and

$$\mathbb{E}\tau \leq \left\lceil \frac{n + m \log_2(d/(d-1))}{\varepsilon_r} \right\rceil + \frac{1}{2^{\varepsilon_r} - 1}. \quad (56)$$

The expected number of fair bits is $r\mathbb{E}\tau$. In particular, if $A_d \leq 2^{k-1}$, taking $r = \lceil \log_2 A_d + 1 \rceil$ gives

$$\mathbb{E}[\text{fair bits}] = O((n + m/d) \log(d+1)). \quad (57)$$

The redraw map may change adaptively with the entire past, but must be fixed before U_t is read.

Messner and Thierauf’s *Modified-Search* instead redraws a violated k -clause uniformly from its $2^k - 1$ satisfying local assignments and obtains the corresponding $c_d \leq 2^k - 1$ condition under their lopsided degree convention (Messner and Thierauf, 2012). Corollary 4.4 addresses a different resource question: the adaptive redraw may use only 2^r outcomes, and hence only $r = O(\log d)$ fresh fair bits, without requiring that the violated local assignment be excluded. Csóka et al. (2022) obtain a constant expected total number of random bits on uniformly subexponential-growth dependency graphs by reusing bits across distant parallel repairs. The present result instead uses fresh conditional entropy, applies to arbitrary bounded-degree graphs, and keeps the original sequential repair map; it does not capture that reuse.

Corollary 4.5 (A Santha–Vazirani tape). *Let the bits (U_j) satisfy, for some $0 < \alpha \leq 1/2$,*

$$\alpha \leq \mathbb{P}\{U_j = 1 \mid U_1, \dots, U_{j-1}\} \leq 1 - \alpha$$

at every positive-probability history (Santha and Vazirani, 1986). Feed successive blocks of r bits through the adaptive injective maps in Corollary 4.4. If

$$\varepsilon_\alpha = r \log_2 \frac{1}{1 - \alpha} - \log_2 A_d > 0,$$

then (53) and (56) hold with ε_α in place of ε_r . Taking $r = k$ and E_t to be the identity runs ordinary clause resampling directly from the Santha–Vazirani tape, without an extractor.

Proposition 4.6 (A one-step min-entropy obstruction). *Let $d \geq 2$ and $k \geq 2$. If $d + 1 \leq 2^{k-1}$, there is a satisfiable k -CNF of dependency degree d whose repair scopes overlap without being equal. From a specified initial assignment, uniform independent k -bit redraws satisfy*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \frac{d+1}{2^k}, \quad (58)$$

$$\dim_H^P \mathcal{N} = \sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \frac{\log_2(d+1)}{k}. \quad (59)$$

There is also a fixed clause-wise redraw source of conditional block min-entropy $\log_2(d+1)$ on which every run is infinite. The dependency graph of this instance is K_{d+1} , for which $\kappa_{K_{d+1}} = d+1$. Thus Theorem 4.1 is sharp for this graph: strict inequality above $\log_2(d+1)$ gives exponential termination, whereas equality can sustain an infinite run. Among guarantees that use only the maximum degree d , the one-step sufficient and obstructing thresholds are separated by less than

$$\log_2 c_d - \log_2(d+1) < \log_2 e$$

bits. When $d+1 = 2^r$, the infinite-run source is an injective r -fair-bit redraw. This last source need not be Santha–Vazirani, so the proposition does not claim sharpness for Corollary 4.5.

Theorem 4.7 (Tree formulas approach the max-degree boundary). *Fix integers $2 \leq d \leq k$. For every height $\ell \geq 1$, there is a satisfiable k -CNF $\Phi_{d,k,\ell}$ whose clause-overlap graph is the full rooted $(d-1)$ -ary tree $T_{d,\ell}$ of height ℓ . Every variable occurs in at most two clauses, every clause overlaps at most d others, and every pair of adjacent bad events is disjoint. Draw the initial assignment uniformly, and then use independent uniform redraws; let P denote this sequential tape law. For any deterministic current-assignment selector,*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = 2^{-k} \kappa_{T_{d,\ell}}, \quad (60)$$

$$\dim_H^P \mathcal{N} = \frac{\log_2 \kappa_{T_{d,\ell}}}{k}. \quad (61)$$

If the selector is computable, the right side of (61) is also the maximal effective strong dimension. Moreover,

$$\kappa_{T_{d,\ell}} \uparrow \frac{d^d}{(d-1)^{d-1}} = c_d \quad (\ell \rightarrow \infty). \quad (62)$$

This tree threshold is the classical regular-tree boundary for the independence polynomial (Scott and Sokal, 2005; Harvey et al., 2018). Thus the max-degree dimension bound in Theorem 4.1 is best possible in the supremum over finite instances, already for tree dependency graphs.

For each fixed ℓ and each fixed selector in the theorem, put $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$. There are a probability law ν_{δ_ℓ} supported on $\mathcal{N}$ and a finite D_ℓ such that

$$\nu_{\delta_\ell}[w] \leq D_\ell P[w]^{\delta_\ell} \quad (63)$$

for every finite tape prefix w . Consequently the same law satisfies $\nu_{\delta_\ell}[w] \leq D_\ell P[w]^s$ for every $0 \leq s \leq \delta_\ell$. Hence the global cylinder threshold (46) is sharp, including at the critical power. This statement does not assert a matching one-step conditional min-entropy law.

The clique and tree lower families are extremal. The clique has partially overlapping scopes and gives the exact one-step threshold for its graph; the trees make the maximum-degree dimension and global source threshold sharp. For general degree- d repairs with $d+1 \leq 2^{k-1}$, the present one-step lower and upper thresholds are $\log_2(d+1)$ and $\log_2 c_d$ bits. Because both lower families are extremal, a matching lower bound for nonextremal overlapping repairs remains open.

These are robustness results for the original local repair map, not stronger satisfiability criteria. The trace form gives the sharp graph-specific upper bound, while Theorem 4.3 trades at most $\log_2(5/4)$ additional bits relative to the worst-case max-degree bound for an explicit prefactor and the end-to-end randomness estimate. Proposition 4.6 also shows why logarithmic fresh-block randomness cannot be removed in that model. Randomness reuse and derandomization can have different guarantees. Viewing the same trace bound across source powers also yields the dimension and pointwise-localization statements. For uniform fixed-arity redraws this is equivalent to rescaling the $s = 1$ history count. The disagreement process in Section 3 shows that this reduction to power one fails for nonuniform state-dependent repair.

Corollary 4.8 (Rainbow perfect matchings). *Let $n \geq 2$. Color the edges of K_{2n} so that each color occurs on at most $q \geq 2$ edges, and run Kolmogorov's perfect-matching repair oracle from an arbitrary perfect matching. A flaw is a pair of vertex-disjoint edges of the same color. Put*

$$A = (2n-1)(2n-3), \quad C = (2n-1)(q-1).$$

Suppose, for some $0 \leq h \leq \log_2 A$, every repair choice has conditional atom probability at most 2^{-h} at every feasible positive-probability history. If, for some $\varepsilon > 0$,

$$h \geq \log_2 \left(\frac{256}{27} C \right) + \varepsilon,$$

then every deterministic nonanticipating selector satisfies

$$\mathbb{P}\{\tau \geq T\} \leq (2n-1)!! e^{n/6} 2^{-\varepsilon T}. \quad (64)$$

For uniform repair choices this is nonvacuous when $q-1 < 27(2n-3)/256$. Writing P for the uniform oracle-action law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2((256/27)(2n-1)(q-1))}{\log_2((2n-1)(2n-3))} \right\}. \quad (65)$$

The same bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff},P}(\omega)$ under the computability assumptions of Corollary 2.4.

The matching oracle is a non-product-space example: its action count, atomicity, and strong commutativity are established by [Kolmogorov \(2018\)](#). The dimensions in (65) refer to its uniform action-outcome tree. The derivations of both applications are given in [Appendix B.5](#).

4.1 Relation to earlier work and scope

Action-level commutativity for resampling oracles was developed by [Kolmogorov \(2018\)](#); the matrix formulation of [Harris et al. \(2025\)](#) is broader. At any fixed power, scalar normalization and a killed-process construction recover the state-history special case of the exact-prefix estimate from their matrix history-DAG lemma. The extension used here is the direct full-prefix formulation, which gives weak-source and pointwise bounds across source powers while permitting colliding actions and selectors that remember action labels. [Corollary 3.2](#) shows that this distinction can be decisive: two implementations with one common source and identical ordinary repair matrices can have nearly opposite dimension and weak-source behavior because an m -way action collision is invisible at power one. A fixed- s normalization can analyze $K_{f,s}$ once that matrix is supplied; it cannot reconstruct $K_{f,s}$ from $K_{f,1}$. The paper does not claim a stronger universal satisfiability threshold.

[Harris and Srinivasan \(2017\)](#) study Rényi entropy of intermediate and terminal Moser–Tardos output distributions, principally for enumerative applications. Here the powered probabilities belong instead to surviving *input-tape cylinders*, usually at powers below one, and describe continued execution and the geometry of exceptional tapes. Neither object determines the other. For extremal partial rejection, confluence and the probability-weighted trace series are established ([Cartier and Foata, 1969](#); [Guo et al., 2019](#); [Jerrum, 2024](#)), and Bernoulli measures on infinite trace boundaries were characterized by [Abbes and Mairesse \(2015\)](#). The finite-state links among weighted growth, Hausdorff dimension, and Kolmogorov complexity are likewise classical in symbolic dynamics and regular infinite languages ([Staiger, 1993](#); [Parry, 1964](#); [Bowen, 1975](#); [Mauldin and Williams, 1988](#); [Simpson, 2015](#)). The present use of this machinery is computational: source powers are inserted into local repair to analyze one fixed algorithm without materializing its full assignment chain.

Reverse reconstruction is basic to entropy compression ([Moser, 2009](#); [Moser and Tardos, 2010](#)). [Messner and Thierauf \(2012\)](#) reconstruct the uniform bits used by satisfiability search and apply Kolmogorov complexity, while [Fialho et al. \(2022\)](#) identify the equality of Moser–Tardos and entropy-compression bounds through the subset-gas cluster expansion. The backward theorem below supplies a nonuniform exact likelihood identity, an arbitrary comparison law, and a relative-entropy formula for its slack.

Finally, the weak-source results keep the selector and available actions fixed; they are narrower than general simulation from weak randomness ([Chor and Goldreich, 1988](#); [Doron et al., 2023](#)). Exact source-power formulas are proved here for disagreement repair and selected extremal families. A compact exact representation for general nonextremal overlapping repair remains open.

5 A backward identity for individual runs

The source-power theorem sums over all live prefixes. A complementary identity controls each realized run without a commutativity or local-lemma assumption.

Let $X_1, \dots, X_n$ be independent finite-valued variables with product law $\pi = \bigotimes_i \pi_i$. Each bad event A depends on coordinates $v(A)$ and has probability $p_A > 0$. Draw $S_0 \sim \pi$. Whenever a bad event is present, a deterministic rule selects A_t , an independent block $Z_t \sim \pi_{v(A_t)}$ is written on its

coordinates, and the resulting assignment is S_t . On the event $E_T = \{\tau \geq T\}$, put

$$\Xi_T = (S_0, Z_1, \dots, Z_T), \quad P_{\text{tape}}(\Xi_T) = \pi(S_0) \prod_{t=1}^T \pi_{v(A_t)}(Z_t).$$

Appendix B.1 gives the formal prefix tree and its source-relative metric.

Let $Y_t = S_{t-1}|_{v(A_t)}$ be the overwritten configuration and let $L_T = (A_1, \dots, A_T)$. Because A_t is true immediately before repair,

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T p_{A_t} \pi_{v(A_t)}(Y_t | A_t). \quad (66)$$

For any subprobability law Q_T that is positive on feasible length- T repair histories, define the backward law

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^T \pi_{v(A_t)}(y_t | A_t). \quad (67)$$

The map

$$\Xi_T \mapsto (S_T, L_T, Y_{1:T})$$

is injective: reverse the repairs, reading each used block from the current assignment and then restoring the overwritten values.

Theorem 5.1 (Backward likelihood identity). *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \quad (68)$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \quad (69)$$

If the instance and $(Q_T)_{T \geq 1}$ are uniformly computable, then

$$-\log_2 P_{\text{tape}}(\Xi_T) - K(\Xi_T | T, \text{instance}) \geq \Delta_T - O(1). \quad (70)$$

The likelihood ratio also gives an exact slack identity. Let $\mathcal{W}_T$ be the prefix-free set of minimal prefixes producing T repairs, let $p_T = P_{\text{tape}}(E_T)$, and put

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)), \quad \mu_T(w) = G_T(R_T(w)), \quad m_T = \sum_{w \in \mathcal{W}_T} \mu_T(w).$$

When $p_T m_T > 0$, define $P_T(w) = P_{\text{tape}}(w)/p_T$ and $\nu_T(w) = \mu_T(w)/m_T$.

Corollary 5.2 (Exact slack and the optimal history law). *For every fixed Q_T ,*

$$-\log_2 p_T = \mathbb{E}_{P_T} \Delta_T + D_2(P_T \| \nu_T) - \log_2 m_T. \quad (71)$$

If $J_T = \sum_{t=1}^T \log_2(1/p_{A_t})$, then

$$\sup_{Q_T} \mathbb{E}[\Delta_T | E_T] = \mathbb{E}[J_T | E_T] - H_2(L_T | E_T), \quad (72)$$

where the supremum is over subprobability laws positive on feasible histories. It is attained by $Q_T(\ell) = \mathbb{P}\{L_T = \ell | E_T\}$ on the conditional support.

Reverse reconstruction and its Kolmogorov-complexity use are established in entropy compression (Moser, 2009; Messner and Thierauf, 2012). The point of Theorem 5.1 is the common nonuniform likelihood: one ratio gives the tail, the tape-by-tape coding saving, and the exact relative-entropy slack. Proofs are in Appendix B.

6 Conclusion

Stopping-time probabilities and exceptional-tape geometry answer different questions. The source-power sum $Z_T(s)$ places them in one family: power one is survival, whereas variation in s controls weak-source robustness and the effective and Hausdorff dimensions of nontermination.

The main theorem makes this family local. When powered matrices for nonadjacent repairs commute, a direct full-prefix argument bounds $Z_T(s)$ by a Cartier–Foata/Shearer trace series, uniformly over deterministic selectors that may use the complete past. This gives source-robust termination and dimension bounds without constructing the full assignment chain. For k -SAT, conditional block min-entropy above $\log_2 \kappa_D$ gives exponential termination; tree and clique formulas establish the stated sharpness regimes, while the recurrent clauses give a pointwise complexity bound for one infinite run.

Exact disagreement repair shows that the full source-power family cannot in general be recovered even from the ordinary repair matrices. Already on P_4 , two implementations driven by one common finite source have identical power-one kernels and the same complete stopping-time law for every selector, yet their exact source-domination thresholds and nontermination dimensions can approach zero and one. At one common domination power, the same imperfect source makes one implementation run forever and gives the other an exponential stopping tail. Thus nominal transition and running-time data can miss almost all of the exceptional-tape geometry and weak-randomness robustness. The backward likelihood identity gives the complementary pathwise result: one exact ratio yields a tail bound, a description-length saving for each surviving prefix, and a relative-entropy formula for the slack.

The limits are equally explicit. The tree family is sharp for dimension and global source domination, not for the one-step weak-source threshold of general nonextremal overlapping repair. For any fixed finite instance, the full assignment gives an exact operator, but usually with exponentially many states. The central remaining problem is to find a natural causal or boundary width that gives a compact exact source-power representation, or to prove a matching hardness boundary for succinct bounded-scope repair.

A Finite-state and tape foundations

Let $\mathcal{G}$ be a finite directed multigraph of nonterminal states with fixed initial state s_0 . Each outgoing edge $e : s \rightarrow t$ represents a distinct next source outcome and has probability $p(e) > 0$. The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path $w = e_1 \cdots e_n$, traversing states $s_0, s_1, \dots, s_n$, determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^n p(e_i).$$

Let $\mathcal{W}_n$ be the set of surviving paths of length n from s_0 . Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies $\max_e p(e) < 1$. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s \rightarrow t} p(e)^\theta, \quad \Lambda(\theta) = \log_2 \rho(B_\theta). \quad (73)$$

For any square matrix C , its spectral radius is $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$. At $\theta = 1$, B_1 is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity $\Lambda(\theta)$ is the base-two pressure, the exponential growth rate of the total θ -weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension; if there is no such component, the nonterminating set is empty.

Use $P[w]$ as the diameter of the cylinder $[w]$. The resulting Hausdorff dimension is the source-relative dimension $\dim_H^P$ defined in the introduction; for a uniform q -ary source it is ordinary Hausdorff dimension in the q -adic metric. For an infinite path $\omega = e_1 e_2 \cdots$, use the corresponding pointwise definition

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define $\dim_{\text{eff}, P}$ by replacing the limsup with a liminf.

Proposition A.1 (Detailed finite-state formula). *There is a unique $\delta \in [0, 1]$ satisfying $\rho(B_\delta) = 1$. Moreover,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \sum_{w \in \mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \quad (74)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \quad (75)$$

$$\dim_H^P \mathcal{N} = \delta. \quad (76)$$

If $B_\delta h = h$ with $h_s > 0$ for every state s , then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \quad (77)$$

is a probability law on surviving paths. For every $w = e_1 \cdots e_n \in \mathcal{W}_n$, define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^n Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \quad (78)$$

For a computable system with computable Q_δ , every nonterminating tape satisfies $\text{Dim}_{\text{eff},P}(\omega) \leq \delta$. A tape that is Martin–Löf random under Q_δ satisfies $\text{dim}_{\text{eff},P}(\omega) = \text{Dim}_{\text{eff},P}(\omega) = \delta$.

The requirement that Q_δ be computable is the effectivity condition needed for the pointwise conclusion. In the strict-interior finite cases below it follows automatically: if the edge probabilities are computable and

$$\rho(B_0) > 1 > \rho(B_1),$$

then $\theta \mapsto \rho(B_\theta)$ is uniformly computable and strictly decreasing, so bisection computes δ . Irreducibility makes the Perron eigenvalue simple; solving $B_\delta h = h$ with $\sum_s h_s = 1$ gives a computable positive eigenvector and hence a computable Q_δ . Endpoint roots used below are explicit.

Proof of Proposition A.1. Since the graph contains a directed cycle, $\rho(B_0) \geq 1$. Since B_1 is substochastic, $\rho(B_1) \leq 1$. If $\theta' > \theta$, then every nonzero entry contributed by an edge is smaller in $B_{\theta'}$ than in B_θ , because $p(e) < 1$. Strict Perron comparison on the common strongly connected support therefore gives $\rho(B_{\theta'}) < \rho(B_\theta)$. Together with continuity, this proves existence and uniqueness of δ . The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1},$$

where e_{s_0} is the coordinate vector of the initial state and $\mathbf{1}$ is the all-ones column vector. Perron–Frobenius theory proves (74); setting $\theta = 1$ gives (75).

If $\theta > \delta$, then $\rho(B_\theta) < 1$, so the total θ -weight of the length- n surviving cylinders tends to zero. These cylinders cover $\mathcal{N}$, giving $\text{dim}_H^P \mathcal{N} \leq \delta$. At $\theta = \delta$, (77) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because h is positive on a finite state space, there is a constant $C < \infty$, independent of w , such that $Q_\delta[w] \leq CP[w]^\delta$. The mass-distribution principle states that a probability measure satisfying this cylinder bound forces $\text{dim}_H^P \mathcal{N} \geq \delta$. It therefore gives the required lower bound, proving (76) and (78) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system with computable Q_δ , coding under Q_δ gives, for every surviving prefix w of length n ,

$$K(w) \leq -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a Q_δ -Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015). $\square$

Under a uniform q -ary source, let $M = (M_{st})$, where M_{st} counts the source symbols that move s to t . Then $B_\theta = q^{-\theta} M$. If

$$\alpha = -\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}$$

is the survival exponent, then

$$\text{dim}_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing $h_{\text{surv}} = \log_2 \rho(M)$ for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\text{surv}}, \quad \dim_H \mathcal{N} = \frac{h_{\text{surv}}}{\log_2 q}.$$

Thus α is the entropy deficit imposed by survival, not an entropy function. Equation (78) is the finite-state form of the comparison used in Theorem 5.1.

B Proofs of the main results

B.1 The consumed-tape model

The tape is the rooted tree of source choices actually queried by the computation. A finite node is a feasible transcript $w = (s_0, z_1, \dots, z_t)$. The root branches to s_0 with probability $\pi(s_0)$. If w is live and the rule next selects A_{t+1} , its children are indexed by local blocks z , with edge probability $\pi_{v(A_{t+1})}(z)$. Hence

$$P_{\text{tape}}[w] = \pi(s_0) \prod_{j=1}^t \pi_{v(A_j)}(z_j). \quad (79)$$

Terminal transcripts are leaves, and infinite branches are exactly the nonterminating consumed tapes. For two distinct infinite branches, let $\omega \wedge \eta$ be their longest common consumed prefix and set

$$d_P(\omega, \eta) = P_{\text{tape}}[\omega \wedge \eta]. \quad (80)$$

All dimension statements assume that these cylinder masses tend to zero along every infinite branch.

Fixed computable enumerations of the outcome alphabets give a canonical self-delimiting encoding of each transcript. The dimensions are unchanged under a computable prefix-tree isomorphism Ψ with computable inverse and bounded source distortion

$$C^{-1} P_{\text{tape}}[w] \leq P'_{\text{tape}}[\Psi(w)] \leq C P_{\text{tape}}[w],$$

because self-information and prefix complexity then change by at most a bounded additive term. No invariance under arbitrary recodings is asserted. A pre-sampled resampling table is only a coupling device: unused entries are excluded, since adjoining independent unused coordinates can change tape dimension without changing the computation.

B.2 Backward likelihood

Proof of Theorem 5.1. Take the ratio of (67) and (66). The final-state factor cancels, as do all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (68).

It remains to use this likelihood ratio. Let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes that produce at least T repairs, and let $\mathcal{R}_T$ be the set of formal records on which G_T is defined. For $w \in \mathcal{W}_T$, write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)).$$

The reconstruction argument shows that R_T is injective.

First observe that G_T has total mass at most one. Indeed, for a fixed list $\ell = (A_1, \dots, A_T)$, the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \leq 1.$$

Define the corresponding mass on live tape prefixes by

$$\mu_T(w) = G_T(R_T(w)), \quad w \in \mathcal{W}_T.$$

Injectivity of R_T gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \leq \sum_{r \in \mathcal{R}_T} G_T(r) \leq 1.$$

Thus μ_T is a subprobability mass function. Its total mass can be strictly less than one because Q_T may itself have mass less than one, or because G_T may assign mass to formal records that are not produced by the algorithm.

For every $w \in \mathcal{W}_T$, the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because $\Delta_T = -\infty$ outside E_T and the cylinders determined by $\mathcal{W}_T$ are disjoint, for every $u \geq 0$,

$$\begin{aligned} \mathbb{P}\{\Delta_T \geq u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} P_{\text{tape}}(w) \\ &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} 2^{-\Delta_T(w)} \mu_T(w) \\ &\leq 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} \mu_T(w) \\ &\leq 2^{-u}. \end{aligned}$$

This proves (69); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Use the canonical effective encoding of the consumed-choice tree from Section B.1, and extend μ_T by zero off $\mathcal{W}_T$. Membership in $\mathcal{W}_T$ is decidable uniformly from (T, w) by simulating the deterministic computation, and $w \mapsto R_T(w)$ is uniformly computable. The finite source factors and Q_T are uniformly computable by hypothesis. Hence $(\mu_T)_{T \geq 1}$ is a uniformly lower semicomputable family of subprobability masses (indeed, uniformly computable). The Kraft–Chaitin coding theorem for lower semicomputable semimeasures, equivalently Shannon–Fano coding in this discrete setting (Shannon, 1948; Fano, 1949; Li and Vitányi, 2019), gives

$$K(\Xi_T \mid T, \text{instance}) \leq -\log_2 \mu_T(\Xi_T) + O(1).$$

The constant is uniform in T and in the realized run. Since

$$-\log_2 \mu_T(\Xi_T) = -\log_2 P_{\text{tape}}(\Xi_T) - \Delta_T,$$

rearranging proves (70). □

Proof of Corollary 5.2. On $\mathcal{W}_T$,

$$\Delta_T(w) = \log_2 \frac{m_T \nu_T(w)}{p_T P_T(w)}.$$

Taking P_T -expectations gives (71). For the second claim, $\Delta_T = J_T + \log_2 Q_T(L_T)$. If r_T is the conditional law of L_T given E_T , then

$$\sum_{\ell} r_T(\ell) \log_2 Q_T(\ell) = -H_2(r_T) - D_2(r_T \| Q_T)$$

when Q_T has total mass one; allocating less mass can only decrease the left side. The maximum is attained at $Q_T = r_T$. $\square$

B.3 Powered matrix theorem

Proof of Lemma 2.1. The (σ, τ) entry of $K_{f,s} K_{g,s}$ is the sum over all action-labelled two-step paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

of

$$(\rho_f(a \mid \sigma) \rho_g(b \mid \sigma'))^s.$$

The diamond bijection maps these paths onto the corresponding g, f paths with the same endpoints and preserves each product before it is powered. The two matrix entries are therefore equal.

For the fixed-arity statement, let $m_f(\sigma, \tau)$ be the number of action labels that move σ to τ when f is addressed. If f is present at σ , then

$$K_{f,s}(\sigma, \tau) = m_f(\sigma, \tau) M_f^{-s} = M_f^{1-s} K_{f,1}(\sigma, \tau).$$

Both sides vanish if f is absent. Hence, for nonadjacent f, g whose ordinary matrices commute,

$$K_{f,s} K_{g,s} = M_f^{1-s} M_g^{1-s} K_{f,1} K_{g,1} = K_{g,s} K_{f,s}.$$

$\square$

Proof of Lemma 2.2. First observe that $K_{H,s}$ is well defined. If x and y are two sources of H , their labels are nonadjacent; otherwise H would contain an edge between them. Their powered matrices commute, and induction on $|H|$ shows that deleting the sources in either order gives the same product.

A stronger conditional statement is proved by induction. Fix an arbitrary past history $\mathfrak{h}$ ending at state σ , and let $M_{\mathfrak{h}}(H)$ be the powered mass of all continuations whose next $|H|$ repairs induce H . The claim is

$$M_{\mathfrak{h}}(H) \leq e_{\sigma}^{\top} K_{H,s} \mathbf{1}. \quad (81)$$

It is immediate for $H = \emptyset$.

Suppose $H \neq \emptyset$, and let f be the flaw selected after $\mathfrak{h}$. If no source of H is labelled f , then $M_{\mathfrak{h}}(H) = 0$. Otherwise there is a unique such source x , since two vertices with the same label are adjacent and cannot both be sources. After an action $a \in \mathcal{A}_f(\sigma)$ leading to $\tau = \Phi_f(\sigma, a)$, the remaining repairs must induce $H - x$. Applying the induction hypothesis separately after every possible action gives

$$\begin{aligned} M_{\mathfrak{h}}(H) &\leq \sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma)^s e_{\Phi_f(\sigma, a)}^{\top} K_{H-x, s} \mathbf{1} \\ &= e_{\sigma}^{\top} K_{f,s} K_{H-x, s} \mathbf{1} = e_{\sigma}^{\top} K_{H,s} \mathbf{1}. \end{aligned}$$

Actions leading to the same state are added in $K_{f,s}$; no recovery of an action from its endpoints is used. The induction is uniform over $\mathfrak{h}$, so the selector may use the complete past action history.

Finally, sum (81) over the initial source symbols. Their powered masses at state σ sum to $b_s(\sigma)$, which gives (14). $\square$

Proof of Theorem 2.3. For every full history DAG H , the definition of $\lambda_f(s)$ gives

$$v_s^\top K_{H,s} \mathbf{1} \leq \left(\prod_{x \in H} \lambda_{L(x)}(s) \right) v_s^\top \mathbf{1} = \prod_{x \in H} \lambda_{L(x)}(s).$$

Since $b_s \leq \gamma_s v_s$ coordinatewise, Lemma 2.2 yields

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq \gamma_s \prod_{x \in H} \lambda_{L(x)}(s). \quad (82)$$

The full history DAG of a flaw word is its labelled dependence graph. Full history DAGs, understood up to the label-preserving isomorphism fixed above, are therefore in one-to-one correspondence with traces; only a subset need be realizable by the algorithm. Summing (82) over realized DAGs and then enlarging to all traces of length T proves

$$Z_T^\Lambda(s) \leq \gamma_s \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_f \lambda_f(s)^{N_f(h)} = \gamma_s a_T(s).$$

The Cartier–Foata identity gives the asserted term-by-term formal-series inequality.

If $q\lambda(s)$ lies in the open Shearer region, the series in (13) converges at $z = q$ and has nonnegative terms. Hence

$$q^T a_T(s) \leq \sum_{t \geq 0} q^t a_t(s) = \frac{1}{P_D(q\lambda(s))},$$

which proves (18).

It remains to derive the local cluster estimate. Assume first that $0 < \theta_s < 1$ and set

$$\bar{\lambda}_f = \frac{\lambda_f(s)}{\theta_s}.$$

By definition of θ_s ,

$$\bar{\lambda}_f \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I) \leq \eta_f.$$

The restricted cluster-expansion enumeration therefore gives, for every independent sink set R ,

$$\sum_{\substack{h \in \mathcal{M}(D) \\ \text{sink}(h)=R}} \prod_f \bar{\lambda}_f^{N_f(h)} \leq \eta(R). \quad (83)$$

This is the standard trace-DAG form of the cluster bound; see Harris et al. (2025, Proposition 3.9). Restricting the sum to traces of length T and restoring the factor θ_s^T gives

$$a_T(s) \leq \theta_s^T \sum_{R \in \text{Ind}(F)} \eta(R).$$

Together with (16), this proves (19). If $\theta_s = 0$, then every $\lambda_f(s) = 0$, so the assertion is immediate for $T \geq 1$; at $T = 0$ it follows from $\sum_\sigma b_s(\sigma) \leq \gamma_s$. $\square$

Proof of Corollary 2.4. The depth- T cylinders indexed by $\mathcal{L}_T^\Lambda$ cover $\mathcal{N}$ and have source diameters at most a fixed initial factor times $\rho_{\max}^T$. By (20),

$$\sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s \leq C_s \vartheta_s^T \longrightarrow 0.$$

This proves that the source-relative s -dimensional Hausdorff measure of $\mathcal{N}$ is zero.

For the effective statement, define on $\mathcal{L}_T^\Lambda$

$$\kappa_{T,s}(w) = \frac{P_{\text{tape}}[w]^s}{\widehat{C}_s \widehat{\vartheta}_s^T}.$$

These are subprobability masses because

$$\sum_{w \in \mathcal{L}_T^\Lambda} \kappa_{T,s}(w) \leq \frac{C_s \vartheta_s^T}{\widehat{C}_s \widehat{\vartheta}_s^T} \leq 1.$$

They are uniformly computable under the stated assumptions. Prefix-free coding gives

$$\begin{aligned} K(w \mid T, \text{instance}) &\leq -\log_2 \kappa_{T,s}(w) + O(1) \\ &= s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1). \end{aligned}$$

Removing the condition T costs $O(\log T)$. Since $\rho_{\max} < 1$, the source self-information of a depth- T prefix grows at least linearly in T . Divide by this self-information and take a limsup; the term $T \log_2 \widehat{\vartheta}_s$ is nonpositive, and the result follows. $\square$

Proof of Corollary 2.5. The cylinders indexed by $\mathcal{L}_T^\Lambda$ are disjoint. Hence

$$\mathbb{P}_\nu\{\tau \geq T\} = \sum_{w \in \mathcal{L}_T^\Lambda} \nu[w] \leq D \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s.$$

Apply (20). $\square$

B.4 Exact disagreement repair

Proof of Theorem 3.1. Let $M(x) = \sum_{v \in V} x_v$. Every selected edge contains one zero and one one. Symbols in $\mathcal{A}_-$, in $\mathcal{A}_p \cup \mathcal{A}_r$, and in $\mathcal{A}_+$ change M by $-1, 0, +1$ and have total source-power weights D_s, N_s, U_s . Since G is connected, a state is live exactly when $1 \leq M \leq N-1$. Conditioning on the next symbol and inducting on T proves (28), even when the selector uses the entire state and action-label history.

The matrix R_s is diagonally similar to the symmetric tridiagonal matrix with diagonal N_s and off-diagonal $(D_s U_s)^{1/2}$. Its eigenvalues are

$$N_s + 2(D_s U_s)^{1/2} \cos \frac{j\pi}{N}, \quad j = 1, \dots, N-1.$$

This proves (29). At $s = 1$,

$$\varrho_1 = N_1 + 2\sqrt{D_1 U_1} c_N < N_1 + D_1 + U_1 = 1.$$

As $s \downarrow 0$, $D_s \rightarrow |\mathcal{A}_-|$, $N_s \rightarrow |\mathcal{A}_p| + |\mathcal{A}_r|$, and $U_s \rightarrow |\mathcal{A}_+|$. Hence $\varrho_s > 1$ near zero. Every source atom lies strictly between zero and one, so ϱ_s is continuous and strictly decreasing. The process therefore terminates exponentially fast and the critical power Δ is unique.

For the lower bound matching the dimension upper estimate, put

$$h_s(k) = \left(\frac{D_s}{U_s}\right)^{k/2} \sin \frac{k\pi}{N}, \quad 1 \leq k < N,$$

and set $h_s(0) = h_s(N) = 0$. Then $R_s h_s = \varrho_s h_s$. At $s = \Delta$, if action a has reference probability $\pi(a)$ and changes the Hamming weight by $d(a) \in \{-1, 0, 1\}$, define

$$\nu_\Delta(a \mid k, z) = \pi(a)^\Delta \frac{h_\Delta(k + d(a))}{h_\Delta(k)}. \quad (84)$$

The eigenvector equation says that these probabilities sum to one. Boundary moves receive probability zero, so ν_Δ is supported on $\mathcal{N}$. For every prefix w in its support, ending at weight k_T ,

$$\nu_\Delta[w] = P[w]^\Delta \frac{h_\Delta(k_T)}{h_\Delta(k_0)}. \quad (85)$$

The ratio is bounded above and below on $\{1, \dots, N-1\}$. The source-relative mass-distribution argument gives the lower Hausdorff bound; (28) at every $s > \Delta$ gives the upper bound. Under computability, a tape random for ν_Δ has effective strong dimension at least Δ by Levin–Schnorr. Since R_Δ is similar to a symmetric matrix with spectral radius one, its powers are bounded; the corresponding critical-power coding masses give the reverse effective bound.

Finally, if (32) holds, disjointness of the live cylinders and (28) give (33). Equation (85) gives the required critical law and domination constant. Thus the same Δ is the exact global source-power threshold and proves all remaining claims. $\square$

Proof of Corollary 3.2. For map H in (36),

$$D_s^H = q^s, \quad N_s^H = m^{1-s}q^s + t^s, \quad U_s^H = t^s.$$

For map L ,

$$D_s^L = m^{1-s}q^s, \quad N_s^L = q^s + t^s, \quad U_s^L = t^s.$$

Since $c_4 = 1/\sqrt{2}$, Theorem 3.1 gives (37) and (38). At $s = 1$ both rules assign masses (q, q, t, t) to $(00, z, 11, \bar{z})$ for every current oriented word z . Thus $K_{f,1}^H = K_{f,1}^L$ for every flaw and state. Their common matrix R_1 also makes the complete stopping-time law identical for every selector by (28); this remains true when the selector sees action labels, because the recurrence is independent of the selected edge and the earlier labels.

For every fixed $s > 0$, as $t \downarrow 0$ one has

$$\varrho_H(s) \longrightarrow m^{1-s}2^{-s}, \quad \varrho_L(s) \longrightarrow 2^{-s}.$$

The first limiting function has unique root $\log m / \log(2m)$. Strict monotonicity and continuity therefore give the first limit in (39). For the second, fix any $s_0 > 0$. Since $2^{-s_0} < 1$, one has $\varrho_L(s_0) < 1$ for all sufficiently small t , and hence $\Delta_L < s_0$. Thus $\Delta_L \rightarrow 0$. Choose m so that $\log m / \log(2m) > 1 - \varepsilon$, and then choose a positive rational t small enough to obtain (34). The source and both maps are then computable, so Theorem 3.1 also gives attainment of the maximal effective strong dimensions.

Now fix $\Delta_L < s_0 < \Delta_H$. The critical law for H satisfies $\nu_H[w] \leq D_H P[w]^{\Delta_H} \leq D_H P[w]^{s_0}$ for every finite action word and is supported on $\mathcal{N}_H$. Feed this same measure on the common tape space to L . The global-domination part of Theorem 3.1 gives L an exponential stopping tail. Finally, from an initial assignment of Hamming weight one, x_1^∞ is neutral for H and immediately decreases to zero for L , whereas y^∞ has the opposite behavior. This proves the language and robustness separations. $\square$

B.5 Weak-randomness applications

For the variable-model calculation, take

$$v_s(\sigma) = \frac{\pi(\sigma)^s}{\sum_{\xi \in \Omega} \pi(\xi)^s}.$$

For $S = v(A)$ and a fixed terminal assignment τ ,

$$\begin{aligned} \sum_{\sigma} v_s(\sigma) K_{A,s}(\sigma, \tau) &= v_s(\tau) \sum_{b \in A} \pi_S(b)^s \\ &= v_s(\tau) r_A(s). \end{aligned}$$

This proves (42).

Proof of Theorem 4.1. For uniform k -clause resampling, $\lambda_A(s) = 2^{-ks}$ for every clause A . A fixed initial assignment and the uniform choice of v_s give $\gamma_s = 2^n$. Therefore Theorem 2.3 gives

$$Z_T^\Lambda(s) \leq 2^n 2^{-ksT} |\{h \in \mathcal{M}(D) : |h| = T\}|.$$

For every $\xi > \kappa_D$, the Cartier–Foata series at $z = \xi^{-1}$ gives

$$|\{h \in \mathcal{M}(D) : |h| = T\}| \leq \frac{\xi^T}{P_D(\xi^{-1}\mathbf{1})}.$$

Multiplying by D_0 proves (45). A conditional atom bound implies $\nu[w] \leq 2^{-hT} = P[w]^{h/k}$ by the chain rule, so this is the special case $s = h/k$ and $D_0 = 1$.

For every $s > \log_2(\kappa_D)/k$, choose $\xi \in (\kappa_D, 2^{ks})$. The resulting source-power sum decays exponentially, which proves the Hausdorff and, under computability, effective dimension bounds by Corollary 2.4. Finally, the symmetric form of Shearer’s theorem (Shearer, 1985; Scott and Sokal, 2005) states that a graph of maximum degree $d \geq 2$ has $z\mathbf{1}$ in its open Shearer region whenever

$$0 \leq z < \frac{(d-1)^{d-1}}{d^d}.$$

Hence $R_D \geq (d-1)^{d-1}/d^d$ and $\kappa_D \leq c_d$. $\square$

Proof of Corollary 4.2. Fix a nonempty $R \subseteq F$. If $\text{Rec}(\omega) = R$, every clause outside R is repaired only finitely often. After a finite tape prefix, the run therefore repairs only clauses in R . Fix such a prefix u . On a history of the instance that retains only the clauses in R , prepend u and apply the original selector whenever it chooses a currently violated clause in R ; otherwise choose the first currently violated clause of R in a fixed order. This defines a deterministic nonanticipating selector for the reduced instance and agrees with the original run on every continuation after u whose later repair labels all lie in R . It is computable whenever the original selector is. The proof of

Theorem 4.1 now applies to the induced graph $D[R]$. For every $s > \log_2(\kappa_{D[R]})/k$, the source-power sum of the suffix decays exponentially.

There are countably many finite prefixes after which the run remains in R . Prefixing rescales the source metric by a constant, and Hausdorff dimension is stable under countable unions. This proves the setwise bound. In the computable case, a fixed prefix and the finite set R cost only an additive constant in a program description. Prefix-free coding of the same suffix bound and passage to the infimum over s prove (50). $\square$

Proof of Theorem 4.3. For a violated k -clause under the uniform reference source, precisely one local assignment is bad, so its powered charge is 2^{-ks} . Take v_s uniform on the 2^n assignments. A fixed initial assignment gives $\gamma_s = 2^n$. Set $\eta_A = \eta = 1/(d-1)$. A clause is adjacent to every other member of its inclusive neighborhood. An independent subset of that neighborhood therefore either consists of the clause itself or is contained in its at most d neighbors. Hence

$$\sum_{I \in \text{Ind}(\Gamma(A))} \eta(I) \leq \eta + (1 + \eta)^d.$$

Consequently

$$\theta_s \leq 2^{-ks} \frac{\eta + (1 + \eta)^d}{\eta} = 2^{-ks} \left(1 + \frac{d^d}{(d-1)^{d-1}} \right) = 2^{-ks} A_d,$$

whereas

$$\sum_{R \in \text{Ind}(F)} \eta(R) \leq (1 + \eta)^m = \left(\frac{d}{d-1} \right)^m.$$

The atom condition implies that every length- T redraw word has probability at most 2^{-hT} . With $s = h/k$, this equals its uniform reference probability raised to s . If $h = \log_2 A_d + \varepsilon$, apply Corollary 2.5 to obtain (53).

Taking $h = k$ gives (54). For the dimension statement, every $s > \log_2 A_d/k$ makes $\theta_s < 1$. Corollary 2.4 applies for every such s ; let s decrease to the displayed value. The bound by one is automatic. $\square$

Proof of Corollary 4.4. Conditioned on the past, injectivity of E_t makes every supported redraw have probability exactly 2^{-r} . The encoder is chosen before U_t , so its dependence on the past does not change this fact. Apply Theorem 4.3 with $h = r$. For a bound $\mathbb{P}\{\tau \geq T\} \leq C_0 2^{-\varepsilon T}$, bound the first $\lceil \log_2(C_0)/\varepsilon \rceil$ terms of $\mathbb{E}\tau = \sum_{T \geq 1} \mathbb{P}\{\tau \geq T\}$ by one and sum the remaining geometric series. With $C_0 = 2^n(d/(d-1))^m$ this gives (56); every repair consumes exactly r bits.

If $A_d \leq 2^{k-1}$ and $r = \lceil \log_2 A_d + 1 \rceil$, then $r \leq k$, $\varepsilon_r \geq 1$, and $r = O(\log(d+1))$. Moreover, $\log_2(d/(d-1)) = O(1/d)$ for $d \geq 2$. Substitution in (56) proves (57). $\square$

Proof of Corollary 4.5. Condition on the entire past before a block is read. The chain rule and the Santha–Vazirani inequalities give, for every $z \in \{0, 1\}^r$,

$$\mathbb{P}\{(U_{j+1}, \dots, U_{j+r}) = z \mid U_1, \dots, U_j\} \leq (1 - \alpha)^r.$$

After an injective map, the same upper bound holds for every redraw atom. Thus Theorem 4.3 applies with $h = r \log_2(1/(1 - \alpha))$. $\square$

Proof of Proposition 4.6. Put $\ell = \lceil \log_2(d+1) \rceil$ and choose distinct words $u_1, \dots, u_{d+1} \in \{0, 1\}^\ell$. Use ℓ common variables and, for each i , a disjoint private block P_i of $k - \ell$ variables. Let C_i be the unique clause on the common variables and P_i that is false when the common word is u_i and P_i is

all zero. The clauses share the common variables, so their dependency graph is K_{d+1} , but their scopes are not equal. Distinct clauses cannot be false simultaneously. The formula is satisfiable, for example by setting one private variable in each P_i to one.

Start with every private block zero and the common word equal to u_1 . Call i active when P_i is still zero and let q be the number of active indices. In every live state the common word is u_i for a unique active i , so C_i is the only violated clause. Write $M = 2^{k-\ell}$. When C_i is repaired, exactly q of its 2^k redraws leave q unchanged: the new private block is zero and the new common word is u_j for an active j . Exactly $(q-1)(M-1)$ redraws move to $q-1$: the new private block is nonzero and the new common word is u_j for one of the other active indices. Every other redraw terminates. Thus the all-power live matrix on $q = 1, \dots, d+1$ is bidiagonal, with

$$R_s(q, q) = q 2^{-ks}, \quad R_s(q, q-1) = (q-1)(M-1)2^{-ks}. \quad (86)$$

Its spectral radius is $(d+1)2^{-ks}$. Proposition A.1 therefore gives (58) and (59), including pointwise attainment.

At $q = d+1$, instead redraw the selected scope uniformly from the $d+1$ assignments $(u_j, 0)$, $1 \leq j \leq d+1$. This source has conditional block min-entropy $\log_2(d+1)$ and remains at $q = d+1$ forever. If $d+1 = 2^r$, the redraw is the image of an injective map applied to r fresh fair bits. Finally,

$$\frac{c_d}{d+1} = \frac{d}{d+1} \left(1 + \frac{1}{d-1}\right)^{d-1} < e,$$

which proves the stated gap. $\square$

Proof of Theorem 4.7. Orient every edge of $T_{d,\ell}$ away from the root and introduce one Boolean variable X_e for each edge. For a vertex v , form a clause C_v containing one literal for each incident edge, using opposite signs at the two endpoints. Add $k - \deg(v)$ private variables to C_v , each used in no other clause. This is possible because $\deg(v) \leq d \leq k$. Every clause has exactly k distinct variables, and every variable occurs once or twice.

Two clauses share a variable exactly when their vertices are adjacent. Their bad events prescribe opposite values to the shared edge variable and are therefore disjoint; nonadjacent clauses have disjoint scopes. The formula is satisfiable. For example, use the sign convention in which the parent literal on an edge is X_e and the child literal is $\neg X_e$, set every $X_e = 0$, and set one private literal of the root to true. Every nonroot clause is satisfied by its parent-edge literal, while the root has a private literal because its degree is $d-1 < k$.

Every bad event is atomic and has probability 2^{-k} . Let

$$N_T(D) = |\{h \in \mathcal{M}(D) : |h| = T\}|.$$

Let n be the number of variables and fix the selector. Every depth- T prefix has probability $2^{-(n+kT)}$. Fix a trace H and a satisfying terminal assignment y . Reconstruct the prefix by induction on $|H|$. For $H = \emptyset$, the reconstructed state is y . Otherwise choose a source x of H , first reconstruct $H - x$, and then restore on the scope of $L(x)$ the unique local assignment that makes $L(x)$ bad; atomicity gives uniqueness. The redraw block associated with x is the restriction of the reconstructed post-repair assignment to the scope of $L(x)$; hence both the predecessor state and the corresponding tape symbol are determined. The result does not depend on the chosen source. Any two source-removal orders are connected by swaps of adjacent incomparable nodes, and incomparable nodes have nonadjacent labels and hence disjoint scopes. Their reverse restorations, including the recorded redraw blocks, therefore commute.

Inductively, the bad events in the reconstructed initial state are exactly the labels of the sources of H . The restored event $L(x)$ is bad; every adjacent bad event is absent by extremality, while every nonadjacent event is unchanged and is governed by the induction hypothesis for $H - x$. Nodes with the same label are comparable, so each source label occurs only once. The selector therefore chooses the label of a unique source node. Removing that node and repeating produces a unique selector-compatible linear extension of H , and hence exactly one live prefix ending at y . Different traces give different repair traces. Conversely, a trace and a terminal assignment determine at most one live prefix, and there are at most 2^n terminal assignments. This is the elementary finite-horizon form of the extremal probability-weighted trace identity (Guo et al., 2019; Jerrum, 2024). Hence, for every $s > 0$,

$$2^{-sn}2^{-ksT}N_T(T_{d,\ell}) \leq Z_T^\Lambda(s) \leq 2^{(1-s)n}2^{-ksT}N_T(T_{d,\ell}). \quad (87)$$

The limit of the trace growth exists without an aperiodicity assumption. Choose one canonical word for each trace. Splitting the canonical word of a length- $(m+n)$ trace after m letters maps it injectively to a pair of traces of lengths m and n : the pair determines its product and hence the original trace. Therefore

$$N_{m+n}(D) \leq N_m(D)N_n(D).$$

Fekete's lemma applied to $\log N_T(D)$ shows that $N_T(D)^{1/T}$ has a limit, whose value is κ_D by (43). It follows from (87) that

$$\lim_{T \rightarrow \infty} Z_T^\Lambda(s)^{1/T} = 2^{-ks} \kappa_{T_{d,\ell}}.$$

At $s = 1$ this proves (60). The unique power at which the rate equals one is $\log_2(\kappa_{T_{d,\ell}})/k$. Applying Proposition A.1 to the reachable live components gives (61) and, for a computable selector, effective attainment.

It remains to calculate the trace growth. Put $b = d - 1$ and define

$$r_0(z) = z, \quad r_{j+1}(z) = \frac{z}{(1 - r_j(z))^b}.$$

As long as the denominators are positive, $r_j(z)$ is the signed root-occupation ratio for a full b -ary subtree of height j . The first positive zero $z_{d,\ell}$ of $P_{T_{d,\ell}}(z\mathbf{1})$ is characterized by $r_\ell(z_{d,\ell}) = 1$. The critical value for the iteration is

$$z_* = \max_{0 \leq r < 1} r(1 - r)^b = \frac{b^b}{(b+1)^{b+1}} = \frac{(d-1)^{d-1}}{d^d}.$$

If $z < z_*$, the iteration increases to the smaller fixed point of $r = z/(1 - r)^b$ and remains below one. If $z > z_*$, there is no fixed point in $[0, 1)$, so the iteration reaches one after finitely many steps. Hence $z_{d,\ell} \downarrow z_*$ and

$$\kappa_{T_{d,\ell}} = z_{d,\ell}^{-1} \uparrow z_*^{-1} = c_d.$$

Finally put $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$. For the fixed selector, consider the finite live chain on full assignments. The exact growth calculation above and the finite-state decomposition give a reachable irreducible component C whose powered transition matrix satisfies $\rho(B_{\delta_\ell, C}) = 1$. Choose a positive right Perron vector h and a live prefix u entering C . On an internal edge $e : x \rightarrow y$ of C , set

$$Q(e \mid x) = P(e)^{\delta_\ell} \frac{h(y)}{h(x)}.$$

Its row sums are one, so prefixing Q by u defines a probability law ν_{δ_ℓ} supported on $\mathcal{N}$. Telescoping along extensions of u , and absorbing the source probability of u and the finitely many ratios of entries of h into D_ℓ , gives (63) for every finite prefix w . Since $P[w] \leq 1$, the same inequality holds with $P[w]^s$ for every $0 \leq s \leq \delta_\ell$. $\square$

Proof of Corollary 4.8. For the matching oracle of Kolmogorov (2018), every flaw has

$$A = (2n - 1)(2n - 3)$$

equiprobable actions, distinct oracle executions have distinct outputs in the action-expanded graph, and the oracle is strongly commutative. Thus its powered charge is A^{-s} . Put $C = (2n - 1)(q - 1)$. The neighborhood of a flaw is covered by four families, one for each endpoint, each containing at most C flaws. Therefore, for constant η ,

$$\sum_{I \in \text{Ind}(\Gamma(f))} \eta^{|I|} \leq (1 + C\eta)^4.$$

Taking $\eta = 1/(3C)$ gives

$$\theta_s \leq \frac{A^{-s}}{\eta} (1 + C\eta)^4 = \frac{256}{27} C A^{-s}.$$

There are at most

$$|F| \leq \frac{n(2n - 1)(q - 1)}{2}$$

flaws, and hence

$$\sum_{R \in \text{Ind}(F)} \eta^{|R|} \leq (1 + \eta)^{|F|} \leq e^{|F|\eta} \leq e^{n/6}.$$

With v_s uniform on the $(2n - 1)!!$ perfect matchings, a fixed initial matching has $\gamma_s = (2n - 1)!!$. The conditional atom bound and the uniform reference oracle give power domination with $s = h/\log_2 A$. Corollary 2.5 now proves (64). Letting s decrease to $\log_2((256/27)C)/\log_2 A$ proves the dimension bound. Under uniform actions, the strict condition $\theta_1 < 1$ is equivalent to $q - 1 < 27(2n - 3)/256$. $\square$

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